MediFlow

Hospital ERP + CRM

Phase 1 MVP — User, Demo & Operations Guide

This document explains what MediFlow is, which module to use when, how to run common workflows, role-based access (RBAC), demo logins, and deployment notes for Vercel demos.

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Repository: https://github.com/rounakdutta121/mediflow

1. What is MediFlow?

MediFlow is a modern single-hospital operations platform that combines front-office CRM (patients, appointments, queue) with clinical EMR basics and OPD billing. It is built with Next.js, MongoDB Atlas, Auth.js, and role-based permissions so each staff role only sees and does what they should.

Phase 1 includes

- Secure login, JWT sessions, RBAC, notifications, audit trail
- Patient registration with UHID, search, and profile
- Departments, doctors/staff directory, doctor schedules
- Appointment booking, check-in, token queue
- Clinical encounters, notes, vitals, diagnoses, signed prescriptions
- Service catalogue, invoices, payments/receipts, auto-invoice from priced Rx on sign
- Operational dashboard, MTD reports, hospital settings
- Light/dark theme, responsive UI, mobile sidebar

Later phases (navigation shells only)

- Admission & Beds
- Laboratory
- Pharmacy
- Insurance / TPA
- Nursing

2. Demo credentials (IDs & passwords)

Use these accounts after seeding the database (`npm run seed`). Passwords are case-sensitive.

Role	Email / ID	Password	Best for
Hospital Admin	admin@mediflow.local	Admin@12345	Full access, settings, staff
Doctor	dr.mehta@mediflow.local	Doctor@12345	EMR, queue, own appointments
Doctor	dr.khan@mediflow.local	Doctor@12345	Second clinician demo
Receptionist	reception@mediflow.local	Reception@12345	Patients, booking, queue, billing
Accountant	accounts@mediflow.local	Accounts@12345	Invoices, payments, reports

Roles defined in code but not seeded by default: NURSE, AUDITOR. Admin can create them under Doctors & Staff.

Demo hospital defaults (seed)

- Hospital name: MediFlow General Hospital
- Hospital code: MFH
- Timezone: Asia/Kolkata · Currency: INR
- Sample patient (if present): Rounak Dutta · UHID MFH-001006
- Sample services: OPD Consultation, Follow-up, ECG, CBC

Security note: Change all demo passwords before any real/ production use. Never commit .env.local.

3. When to use which module

Situation	Open this module	Who usually uses it
New walk-in / registration	Patients !’ Register	Reception
Schedule a visit	Appointments !’ Book / Patient !’ Book	Reception
Patient arrived today	Queue !’ Check-in	Reception / Nurse
Doctor starts consult	Queue !’ Start consult / Clinical !’ New	Doctor
Write Rx / vitals / notes	Clinical !’ Encounter workspace	Doctor
Collect money / OPD bill	Billing !’ Create invoice / Collect payment	Reception / Accounts
Month-to-date KPIs	Reports / Dashboard	Admin / Accounts
Investigate who changed what	Audit	Admin / Auditor
Hospital name, contact, TZ	Settings	Admin only
Add doctor / change role	Doctors & Staff	Admin
Add specialty units	Departments	Admin

4. How to use — step by step

4.1 Sign in

- Open the app URL (local http://localhost:3000 or your Vercel URL).
- Enter email + password from the credentials table.
- Optional: use the theme toggle (sun/moon) for light or dark mode.
- Sidebar only shows modules your role can access.

4.2 Register a patient

- Go to Patients !’ Register patient (needs patients:write).
- Fill first/last name and phone (required). Add allergies/conditions as comma lists.
- Save — system assigns a UHID (e.g. MFH-001007).
- From profile: Book appointment, Start encounter, Create invoice

(only if your role allows).

4.3 Book an appointment

- Appointments !’ Book appointment, or Patient !’ Book appointment.
- Select patient, doctor, department, date, time, type (OPD / follow-up / walk-in / teleconsult).
- Save. Appointment appears on the appointments board for that date.

4.4 Check-in & queue

- Open Queue (queue:manage).
- Check in a scheduled patient for today — token is generated.
- Doctors with clinical:write see Start consult; reception typically only manages queue status.

4.5 Clinical encounter (EMR)

- Clinical !’ New encounter, or Start encounter from queue/patient.
- Document chief complaint, history, examination, assessment, plan, vitals, diagnosis.
- Add prescriptions with optional service/price; signing a priced Rx can auto-create an invoice.
- Edit notes/vitals while open; delete only with clinical:write.

4.6 Billing & payments

- Billing !’ Create invoice: pick patient, add catalogue services, discount, save.
- Open invoice !’ Collect payment (cash/card/UPI etc. as available in form).

- Status moves draft/issued !’ partially paid !’ paid. Balance shows on dashboard.

4.7 Staff, departments, settings

- Doctors & Staff: create users with role + password; edit/deactivate (staff:write = Admin).
- Departments: create/edit/delete active departments (departments:write = Admin).
- Settings: edit hospital name, code, address, phone, email, timezone, currency (settings:manage = Admin).

4.8 Notifications

- Bell icon in the top bar lists targeted alerts (patient, appointment, clinical, billing, system).
- Hospital Admin receives all notifications; others receive role-targeted ones.
- Mark one read, mark all read, or delete from the panel.

5. RBAC — roles & permissions

Every page and API is permission-gated. Missing permission !’ UI hides the action; direct URL redirects to dashboard?error=forbidden; API returns 403.

5.1 Permission catalogue

- dashboard:view — home KPIs
- patients:read / patients:write — view vs register/update/delete
- appointments:read / appointments:write — list vs book/update/cancel
- queue:manage — check-in and queue board
- staff:read / staff:write — directory vs create/edit staff
- departments:read / departments:write — view vs manage departments
- clinical:read / clinical:write — view EMR vs create/edit/sign/delete
- billing:read / billing:write — view invoices vs create/pay/delete
- reports:view — analytics
- audit:view — audit log
- settings:manage — hospital configuration

5.2 Role matrix (who can do what)

Capability	Admin	Reception	Doctor	Nurse	Accounts	Auditor
Dashboard	Yes	Yes	Yes	Yes	Yes	Yes
Patients read	Yes	Yes	Yes	Yes	Yes	Yes
Patients write	Yes	Yes	No	No	No	No
Appointments write	Yes	Yes	No	No	No	No
Queue	Yes	Yes	Yes	Yes	No	No
Clinical write	Yes	No	Yes	No	No	No
Clinical read	Yes	No	Yes	Yes	No	Yes
Billing write	Yes	Yes	No	No	Yes	No
Reports	Yes	No	No	No	Yes	Yes
Audit	Yes	No	No	No	No	Yes
Staff write	Yes	No	No	No	No	No
Settings	Yes	No	No	No	No	No

5.3 Practical RBAC examples

- Reception cannot open /clinical — redirected as forbidden.
- Doctor cannot open /billing — redirected as forbidden.
- Accountant can invoice and view reports, but not manage queue/staff.
- Only Admin edits Settings and creates staff with

roles/ passwords.

6. Recommended demo script (5–10 minutes)

- 1. Login as Admin — show dashboard KPIs, toggle dark mode, open notifications.
- 2. Switch to Reception — register a patient, book appointment with Dr Mehta, check-in on Queue.
- 3. Switch to Doctor — open Queue/ Clinical, start encounter, add vitals +

Rx with price, sign.

- 4. Switch to Accounts — open Billing, confirm invoice / collect payment, open Reports.
- 5. Back to Admin — show Audit trail and editable Hospital settings.

7. Local & Vercel setup

Local

- npm install
- Copy .env.example !’ .env.local and set MONGODB_URI + AUTH_SECRET + AUTH_URL
- npm run seed
- npm run dev !’ http://localhost:3000

Vercel demo

- Import GitHub repo rounakdutta121/mediflow
- Set env: MONGODB_URI, AUTH_SECRET, AUTH_URL (your https://….vercel.app)
- Deploy, then run seed once against the same Atlas URI
- Atlas network access must allow Vercel (0.0.0.0/0 is typical for demos)

Useful commands

- npm run seed — demo hospital, users, departments, services
- npm run build — production build check
- npm run test:rbac — permission matrix against running server
- npm run test:frontend — Playwright UI smoke

8. Support & scope notes

- Phase 1 is single-hospital OPD-focused CRM/ERP foundation.
- Admission, lab, pharmacy, insurance, and nursing are placeholders for later phases.
- Soft-delete is used for many entities (e.g. patients marked inactive via Delete in UI).
- For demos, keep seeded credentials handy; rotate them for any shared public deployment.

End of MediFlow Phase 1 User & Demo Guide
MediFlow — Clinical ops, flowing with clarity.